

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 6

6 Data Science (djw1005)

Here are two binary sequences:

$$x = 111010100100001011110011$$

$$y = 000000011111111111110000$$

fitting models

- (a) Consider a model in which each element of x is an independent Bernoulli random variable, $\text{Bin}(1, p)$. Estimate p , and give a formula for the log likelihood of the x sequence. Repeat for y . Explain why the log likelihoods are the same for the two sequences. [4 marks]

hypothesis test

- (b) The model from part (a) might seem a poor choice for y . Explain how to conduct a hypothesis test to determine whether this is so. In your answer you should define the test statistic you have invented for the test, and you should explain your choice of one-sided or two-sided testing. [8 marks]

Markov chain,
stationary
distribution,
learning a Markov
chain

- (c) Consider an alternative model, in which y is generated from a two-state Markov chain, with the first item of y drawn from the chain's stationary distribution. Give a formula for the log likelihood of y , and explain how to fit the model. [8 marks]

[*Note:* You do not need to find numerical answers. You may be interested to know that for the independent model the log likelihoods are both -16.55 , and for the Markov model the log likelihood of x is -16.47 and that for y is -9.00 .]

Answer:

— *Solution notes* —

Part (a). The log likelihood is

$$\log \Pr(x) = \log \left[\prod_i \begin{cases} p & \text{if } x_i = 1 \\ 1-p & \text{if } x_i = 0 \end{cases} \right] = n_1 \log p + n_0 \log(1-p)$$

where $n_1 = 13$ is the number of ones, and $n_0 = 11$ the number of zeros. This is maximized at $\hat{p} = 13/24$. The answer is exactly the same for y , since it has the same n_1 and n_0 . *Marks: 2 for the log likelihood, 1 for MLE, 1 for comment.*

Part (b). Intuitively, y has fewer flips than we'd expect (it has one flip $0 \rightarrow 1$, and one flip $1 \rightarrow 0$) whereas x has many more. We can use this as the basis for a hypothesis test. If the model is correct we'd expect a certain number of flips; if the model is incorrect then there might be fewer flips or there might be more flips, depending on whether the true model has positive or negative correlations between successive items; and therefore we use a two-sided test.

- (a) Generate many sequences from the fitted model from part (a).
- (b) For each, evaluate the test statistic described above.
- (c) Plot a histogram of these values, and mark on the actual observed value i.e. 2.
- (d) Let p be the two-sided probability of a result as extreme as this or more extreme, which in this case will be twice the fraction of cases with ≤ 2 flips (assuming the observed value is smaller than expected, which is intuitively likely to be the case)
- (e) If $p < 5\%$, reject the null hypothesis i.e. the hypothesis that the fitted model is correct.

Marks: 2 for test statistic, 2 for sidedness, 3 for writing out the procedure, 1 for tying it all together by relating test statistic intuition to p-value computation. There is creativity required to invent a test statistic, and to decide on sidedness. The rest is bookwork.

Part (c). Let the transition matrix be $P_{01} = p$, $P_{10} = q$, $P_{00} = 1-p$, $P_{11} = 1-q$. The stationary distribution is

$$\mathbb{P}(X_0 = 0) = \frac{q}{p+q}, \quad \mathbb{P}(X_0 = 1) = \frac{p}{p+q}$$

(which we can derive from detailed balance, or from full balance). The log likelihood of the full sequence is

$$\begin{aligned} \log \Pr(y) &= \log \left[\frac{q}{p+q} (1-p)^{n_{00}} p^{n_{01}} (1-q)^{n_{11}} q^{n_{10}} \right] \\ &= \log q - \log(p+q) + n_{00} \log(1-p) + n_{11} \log(1-q) + n_{01} \log p + n_{10} \log q \end{aligned}$$

where n_{00} etc. are transition counts, $n_{00} = 9$, $n_{01} = 1$, $n_{11} = 12$, $n_{10} = 1$. To fit this model, simply maximize over $p, q \in [0, 1]$, by simultaneously solving

$$\begin{aligned} \frac{\partial}{\partial q} \log \text{lik} &= \frac{1}{q} - \frac{1}{p+q} - \frac{n_{11}}{1-q} + \frac{n_{10}}{q} = 0 \\ \frac{\partial}{\partial p} \log \text{lik} &= -\frac{1}{p+q} - \frac{n_{00}}{1-p} + \frac{n_{01}}{p} = 0. \end{aligned}$$

Marks: 3 for the equilibrium distribution, 3 for the log likelihood formula, 2 for MLE. A bonus point for describing numerical optimization over a constrained domain.
